

Multirate Stein Variational Gradient Descent for Efficient Bayesian Sampling

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Abstract

Many particle-based Bayesian inference methods use a single global step size for all parts of the update. In Stein variational gradient descent (SVGD), however, each update combines two qualitatively different effects: attraction toward high-posterior regions and repulsion that preserves particle diversity. These effects can evolve at different rates, especially in high-dimensional, anisotropic, or hierarchical posteriors, so one step size can be unstable in some regions and inefficient in others. We derive a multirate version of SVGD that updates these components on different time scales. The framework yields practical algorithms, including a symmetric split method, a fixed multirate method (MR-SVGD), and an adaptive multirate method (Adapt-MR-SVGD) with local error control. We evaluate the methods in a broad and rigorous benchmark suite covering six problem families: a 50D Gaussian target, multiple 2D synthetic targets, UCI Bayesian logistic regression, multimodal Gaussian mixtures, Bayesian neural networks, and large-scale hierarchical logistic regression. Evaluation includes posterior-matching metrics, predictive performance, calibration quality, mixing, and explicit computational cost accounting. Across these six benchmark families, multirate SVGD variants

improve robustness and quality-cost tradeoffs relative to vanilla SVGD. The strongest gains appear on stiff hierarchical, strongly anisotropic, and multimodal targets, where adaptive multirate SVGD is usually the strongest variant and fixed multirate SVGD provides a simpler robust alternative at lower cost.

Keywords: Stein variational gradient descent, Bayesian computation, multirate integration, particle-based variational inference, adaptive time stepping

1. Introduction

Modern Bayesian inference often requires efficient sampling from complex, high-dimensional, and multimodal posterior distributions. As these problems grow in scale and complexity, traditional sampling methods face significant computational and statistical challenges. The advent of automatic differentiation has enabled the widespread use of gradient-based samplers, making methods such as stochastic gradient Langevin dynamics (SGLD) and stochastic gradient Hamiltonian Monte Carlo (SGHMC) practical and popular choices for large-scale applications [1, 2]. While these approaches advance a single Markov chain using noisy gradients, an alternative paradigm is offered by interacting particle methods, which evolve a population of particles whose empirical distribution approximates the target posterior. This population-based view is central to particle-based variational inference [3]. Within this class, SVGD is a prominent deterministic representative, and recent work has examined acceleration, particle degeneracy, and kernel design in related Stein-particle methods [3, 4, 5]. It opens new avenues for improv-

17 ing sampling efficiency and robustness, especially in challenging inference
18 settings.

19 A useful unifying perspective is to view interacting particle methods as
20 discretizations of a particle flow (possibly with a stochastic component) that
21 transports an initial ensemble toward the target distribution [6]. Here, we
22 focus on Stein variational gradient descent (SVGD) [7] as a representative in-
23 teracting particle method. SVGD constructs a velocity field for the particles
24 based on the target density and a reproducing kernel Hilbert space (RKHS)
25 structure, and updates the particles by discretizing this flow [7]. The veloc-
26 ity field naturally separates into an attractive drift, which moves particles
27 toward high-density regions, and a repulsive interaction, which maintains
28 particle diversity and support coverage of multiple modes.

29 This drift-repulsion structure creates the main numerical challenge stud-
30 ied in this paper. When particles crowd, repulsion can become stiff and
31 destabilize updates. On the other hand, the drift can require finer resolution on
32 anisotropic targets to follow the geometry of $\log p$ closely. A single step size
33 must therefore balance competing requirements and can either over-damp
34 repulsion, leading to loss of diversity and mode coverage, or under-resolve
35 drift, leading to slow progress.

36 Moreover, repulsion typically requires kernel evaluation at each step, so
37 increasing the stability by reducing the step-size can become computationally
38 expensive. Finally, repulsive forces can weaken in high dimensions or under
39 limited particle resolution, leading to particle collapse and poor support of
40 multiple modes [4, 8]. In practice, the resulting single-step-size compromise
41 is both statistically and computationally inefficient: it can degrade support

42 coverage and predictive quality while still spending substantial kernel work
43 to stabilize the wrong part of the update.

44 A common approach to solving systems with significantly different com-
45 ponents is to use partitioned methods such as implicit-explicit (IMEX) [9, 10]
46 and multirate integration methods [11]. In the context of ordinary differen-
47 tial equations (ODEs), multirate methods exploit a natural splitting of the
48 dynamics into fast and slow components, allowing each to be integrated with
49 a step size appropriate to its scale [12, 13, 14]. For example, in a split ODE
50 of the form $\dot{x} = \phi_{\text{fast}}(x) + \phi_{\text{slow}}(x)$, multirate schemes apply more frequent,
51 smaller steps to the stiff or rapidly varying part, while updating the slow part
52 less often. This approach improves both stability and efficiency in multi-scale
53 dynamical systems, and has been successfully applied in a variety of scientific
54 computing contexts [10, 15].

55 Motivated by advances in multirate time-integration for multi-scale dy-
56 namical systems—including MrGARK, IMEX, and related splitting schemes
57 [12, 13, 14] we propose to integrate the drift and repulsion terms on sepa-
58 rate time scales. This leads to a number of multirate SVGD variants, in-
59 cluding symmetric splitting, fixed repulsion substepping, and adaptive error-
60 controlled drift substepping.

61 To assess the effectiveness of these methods, we conduct a comprehensive
62 empirical evaluation. We compare our multirate SVGD variants with estab-
63 lished approaches from the literature across a range of benchmark problems,
64 using a variety of task-appropriate metrics. Results highlight the strengths
65 and limitations of each method under diverse geometric and statistical chal-
66 lenges.

67 *Contributions*

68 The main contributions of this work are:

- 69 • Multirate SVGD formulations that explicitly separate drift and repul-
70 sion and support symmetric splitting and fixed multirate schedules.
- 71 • An adaptive error-controlled multirate SVGD variant that selects sub-
72 steps based on a local error estimator while keeping repulsion stable.
- 73 • A broad empirical study showing that multirate SVGD improves ro-
74 bustness and quality-cost tradeoffs compared to SVGD and represen-
75 tative chain methods, with the strongest gains on anisotropic, multi-
76 modal, and hierarchical targets.
- 77 • A benchmark suite spanning many different targets, equipped with
78 appropriate diagnostics, early stopping, performance evaluation metrics
79 for efficiency comparisons.

80 We present our work in the following: Section 2 presents the SVGD vari-
81 ants and the multirate construction. Section 3 describes the benchmarks and
82 experimental protocol and reports results. Section 4 summarizes findings and
83 discusses directions for future work.

84 **2. Method**

85 *2.1. Background: SVGD as a particle flow*

86 Let $p(x)$ denote a target density on $\mathbb{R}^d$ and let $\{x_i^t\}_{i=1}^N$ be a set of par-
87 ticles at iteration t . SVGD evolves particles by following a deterministic

88 interacting-particle flow whose velocity field is constrained to RKHS [7]. A
 89 first-order discretization of this flow is the update

$$x_i^{t+1} = x_i^t + h \phi(x_i^t), \quad (1)$$

90 where $h > 0$ is a step size and the SVGD velocity field is

$$\phi(x) = \frac{1}{N} \sum_{j=1}^N \left[k(x_j, x) \nabla_{x_j} \log p(x_j) + \nabla_{x_j} k(x_j, x) \right]. \quad (2)$$

91 The first term in eq. (2) is an *attractive* drift that pushes particles toward
 92 high-density regions via the score $\nabla \log p$. The second term is a *repulsive*
 93 interaction that discourages particle collapse. The multirate constructions
 94 below are kernel-agnostic: they apply once a kernel-induced attractive field
 95 and repulsive field have been specified.

96 2.2. Drift–repulsion splitting and multirate integration

97 The central numerical observation motivating this work is that the drift
 98 and repulsion components of eq. (2) can impose different stability constraints
 99 and can naturally evolve on different time scales. We therefore write the
 100 SVGD flow as a split system

$$\dot{x} = f_{\text{rep}}(x) + f_{\text{drift}}(x), \quad (3)$$

101 where, for particle i ,

$$f_{\text{drift}}(x_i) := \frac{1}{N} \sum_{j=1}^N k(x_j, x_i) \nabla_{x_j} \log p(x_j), \quad (4)$$

$$f_{\text{rep}}(x_i) := \frac{1}{N} \sum_{j=1}^N \nabla_{x_j} k(x_j, x_i). \quad (5)$$

102 We next define multirate discretizations that apply different substepping
 103 strategies to f_{drift} and f_{rep} while keeping a common macro-step size h .

104 *Strang-split SVGD*. Define the forward Euler update that advance the split
 105 system by a step size τ :

$$\Psi_{\text{rep}}^{\tau}(x) = x + \tau f_{\text{rep}}(x), \quad \Psi_{\text{drift}}^{\tau}(x) = x + \tau f_{\text{drift}}(x).$$

106 *Strang splitting* [16] is a classical second-order method that constructs a
 107 macro-step of size h by composing such substeps. In our setting, one macro
 108 step takes the form

$$x^{t+1} = \Psi_{\text{rep}}^{h/2} \circ \Psi_{\text{drift}}^h \circ \Psi_{\text{rep}}^{h/2}(x^t), \quad (6)$$

109 which alternates half a repulsion step, a full drift step, and another half
 110 repulsion step. In practice, the half-steps can be implemented as $\frac{h}{2m}$ smaller
 111 substeps for stability.

112 *Fixed multirate SVGD (MR-SVGd)*. Fixed multirate integrators provide a
 113 principled way to allocate different step sizes to fast and slow components of
 114 a split system [11, 12, 13]. In our SVGD setting, however, higher-order multi-
 115 rate schemes are often not cost-effective: their additional stages translate into
 116 multiple evaluations of the drift and repulsion fields per macro step, and each
 117 such evaluation is expensive due to gradient computations and, in particular,
 118 $O(N^2)$ kernel interactions. Moreover, practical accuracy is also limited by
 119 finite-particle effects and by choices such as bandwidth selection and early
 120 stopping, which can mute the benefits of increasing time-integration order
 121 at a fixed computational budget. We therefore utilize a first-order multirate
 122 construction using Euler updates that preserves a macro step of size h for
 123 the drift while resolving the repulsion with m substeps:

$$x^{t+1} = \Psi_{\text{drift}}^h \circ \left(\Psi_{\text{rep}}^{h/m} \right)^m (x^t). \quad (7)$$

Algorithm 1 Fixed multirate SVGD (MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^N$, step size h , repulsion substeps m , kernel k

```
1:  $x \leftarrow \{x_i\}$ 
2: for  $s = 1$  to  $m$  do
3:   compute repulsion  $f_{\text{rep}}(x)$  using eq. (5)
4:    $x \leftarrow x + (h/m) f_{\text{rep}}(x)$ 
5: end for
6: compute drift  $f_{\text{drift}}(x)$  using eq. (4)
7:  $x \leftarrow x + h f_{\text{drift}}(x)$ 
8: return  $x$ 
```

124 This retains a coarse drift step while stabilizing the repulsive interaction
125 when particles crowd, at the cost of additional kernel evaluations. The cor-
126 responding implementation-level procedure is summarized in algorithm 1.

127 *Adaptive multirate SVGD (Adapt-MR-SVGD).* Although fixed- m schedules
128 are simple and robust, they impose a uniform drift resolution along the entire
129 trajectory. As a result, they may spend unnecessary computational effort in
130 benign regions while providing insufficient resolution in locally stiff regimes.
131 To address this imbalance, we choose the number of drift microsteps m adap-
132 tively at each macro step, while retaining one repulsion update per macro
133 step. This follows the standard local-error-control perspective used in ODE
134 integrators, where local error estimates are used to adjust resolution [17, Ch.
135 II]. After the repulsion update, denote the state by $x = \Psi_{\text{rep}}^h(x^t)$. We esti-
136 mate the local drift-discretization error by comparing one drift step of size h

137 with two consecutive drift half-steps of size $h/2$,

$$x_{\text{full}} = x + h f_{\text{drift}}(x), \quad (8)$$

$$x_{\text{half}} = x + \frac{h}{2} f_{\text{drift}}(x), \quad (9)$$

$$\tilde{x}_{\text{full}} = x_{\text{half}} + \frac{h}{2} f_{\text{drift}}(x_{\text{half}}). \quad (10)$$

138 We then form the relative discrepancy

$$\rho = \frac{\|\tilde{x}_{\text{full}} - x_{\text{full}}\|_2}{\|x\|_2 + \epsilon}, \quad (11)$$

139 which serves as a dimensionless local error indicator for the drift update.
 140 $\epsilon > 0$ is a small constant to avoid division by zero. For a first-order drift
 141 integrator, this local error behaves as [17, Ch. II.4]:

$$\rho(h) = Ch^2 + \mathcal{O}(h^3),$$

142 where C is the principal error constant dependent on f and its derivatives.
 143 If the drift step is replaced by m microsteps of size h/m , then

$$\rho_m \approx C(h/m)^2 \approx \rho/m^2.$$

144 Imposing the target local error tolerance $\rho_m \approx \text{Tol}$ gives $m \approx \sqrt{\rho/\text{Tol}}$. We
 145 therefore use the clipped integer controller

$$m \leftarrow \text{clip}\left(\left\lceil \sqrt{\rho/\text{Tol}} \right\rceil, m_{\min}, m_{\max}\right), \quad (12)$$

146 which increases m when $\rho > \text{Tol}$ and decreases it when $\rho \leq \text{Tol}$, while
 147 enforcing the robustness bounds $m_{\min} \leq m \leq m_{\max}$. The macro-step update
 148 is then

$$x^{t+1} = \left(\Psi_{\text{drift}}^{h/m}\right)^m \circ \Psi_{\text{rep}}^h(x^t), \quad m = m(x^t, h). \quad (13)$$

149 This strategy increases drift resolution only when the local error indicator
 150 suggests stiffness or rapid variation in the drift component. Algorithm 2
 151 provides the complete algorithm for this method.

152 *Remark 1.* In practice, if $m \leq 2$ we can reuse the already computed, locally-
 153 higher-order solution $\tilde{x}_{\text{full}}$ to avoid extra drift evaluations.

154 3. Experiments and Results

155 We evaluate multirate SVGD variants on six benchmark groups: a 50D
 156 Gaussian target, a suite of 2D synthetic targets, UCI logistic regression, a
 157 2D Gaussian mixture (mix8), a one-hidden-layer Bayesian neural network
 158 (BNN), and a large-scale hierarchical logistic-regression (HLR) benchmark.
 159 The comparison focuses on both statistical quality and computational cost.

160 3.1. Evaluation Metrics

161 We track particle evolution, and in case of chain methods, we use a rolling
 162 window, to calculate performance metrics during the sampling process. We
 163 use the following diagnostics that separately assess distributional fidelity,
 164 predictive quality, and computational cost.

165 *Moment fidelity.* When reference moments are available, we report $e_\mu =$
 166 $\|\hat{\mu} - \mu_{\text{ref}}\|_2$ and $e_\Sigma = \|\hat{\Sigma} - \Sigma_{\text{ref}}\|_F$, where $\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$ and $\hat{\Sigma} = \frac{1}{N} \sum_{i=1}^N (x_i -$
 167 $\hat{\mu})(x_i - \hat{\mu})^\top$. These summarize first- and second-order bias, which is critical
 168 in the Gaussian and synthetic-target settings.

169 *Stein discrepancy.* For distributional fidelity, we use kernel Stein discrepancy
 170 (KSD) with score $s_p(x) = \nabla_x \log p(x)$ and an RBF base kernel $k_h(\cdot, \cdot)$ [18, 19].

171 The corresponding Stein kernel is

$$\begin{aligned}\kappa_p(x, x') &= s_p(x)^\top s_p(x') k_h(x, x') + s_p(x)^\top \nabla_{x'} k_h(x, x') \\ &\quad + s_p(x')^\top \nabla_x k_h(x, x') + \text{tr}(\nabla_x \nabla_{x'} k_h(x, x')) ,\end{aligned}\tag{14}$$

172 and the empirical discrepancy is

$$\text{KSD}(X; p) = \left(\frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \kappa_p(x_i, x_j) \right)^{1/2} .\tag{15}$$

173 Under standard kernel conditions, KSD is zero only at the target law and in-
174 creases under global mismatch, including location bias, covariance distortion,
175 and mode omission [18, 19].

176 *Mean log-density fit.* We complement KSD with the empirical mean log-
177 density

$$\overline{\log p}(X) = \frac{1}{N} \sum_{i=1}^N \log p(x_i),\tag{16}$$

178 which corresponds to the logarithmic scoring rule [20]. This metric quantifies
179 average concentration in high-density regions, but it is local in $\log p(x)$ and
180 therefore does not, by itself, certify global distributional agreement. Report-
181 ing eqs. (15) and (16) separates global fit from local density concentration.

182 *Predictive performance and calibration.* For UCI and BNN tasks, we re-
183 port accuracy, negative log-likelihood $\text{NLL} = -\frac{1}{M} \sum_{m=1}^M [y_m \log \hat{p}_m + (1 -$
184 $y_m) \log(1 - \hat{p}_m)]$, and expected calibration error $\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{M} |\text{acc}(\mathcal{B}_b) -$
185 $\text{conf}(\mathcal{B}_b)|$. This combination separates discrimination (accuracy), probabilis-
186 tic sharpness (NLL), and calibration quality (ECE).

187 *Mode-structure diagnostics.* For structured multimodal targets, we report
 188 mode coverage, normalized mode entropy, and mode imbalance to detect
 189 mode collapse and uneven mass allocation; minimum mode mass is retained
 190 as a secondary check when informative. For the 2D synthetic targets, we
 191 additionally retain grid L_1 discrepancy and moment diagnostics as secondary
 192 spatial mismatch measures.

193 *Cost accounting.* In addition to wall-clock time, we track gradient- and kernel-
 194 evaluation counts to support hardware-agnostic efficiency comparisons across
 195 methods with different per-step workloads.

196 *Mixing efficiency.* To quantify dependence, we report ESS from a one-dimensional
 197 diagnostic sequence $z_1, \dots, z_T$ built from the first coordinate in our imple-
 198 mentation. For particle methods at a checkpoint, z_i is the first coordinate of
 199 particle i ($i = 1, \dots, N$). For single-chain baselines, z_i is the first coordinate
 200 of the i -th state in a rolling window of recent iterates (window length as set
 201 in each benchmark script). Define

$$\bar{z} = \frac{1}{T} \sum_{i=1}^T z_i, \quad \hat{\gamma}_k = \frac{1}{T-k} \sum_{i=1}^{T-k} (z_i - \bar{z})(z_{i+k} - \bar{z}), \quad \hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0}. \quad (17)$$

202 Autocorrelations are evaluated up to $k_{\max} = \min(\lfloor T/2 \rfloor, 100)$. Let K be
 203 the first lag in $\{1, \dots, k_{\max}\}$ such that $\hat{\rho}_K < 0$; if no such lag exists, we set
 204 $K = k_{\max}$. The reported ESS is

$$\text{ESS}(z) = \frac{T}{1 + 2 \sum_{k=1}^K \hat{\rho}_k}. \quad (18)$$

205 This initial-positive truncation controls high-lag noise in the empirical auto-
 206 correlation and provides a stable scalar proxy for mixing. Because eq. (18)

is computed from a single diagnostic coordinate, we interpret ESS as a univariate mixing indicator rather than a full multivariate efficiency estimate. Accordingly, distribution-level conclusions rely primarily on KSD and moment/mode diagnostics, with ESS used as a complementary statistic.

Lower is better for e_μ , e_Σ , KSD, NLL, ECE, and grid L_1 , whereas higher is better for ESS, accuracy, coverage, entropy, minimum mode mass, and mean log-probability.

3.2. Experimental protocol

Across benchmarks, we compare the same method family whenever applicable: vanilla SVGD, Strang-split SVGD, fixed multirate SVGD (MR-SVGD), adaptive multirate SVGD (Adapt-MR-SVGD), and the single-chain baselines SGLD and an SGHMC-style method. The SVGD-family updates are those defined in section 2, namely eqs. (1), (6), (7) and (13).

Remark 2. For the single-chain baselines, we use the standard stochastic-gradient update

$$x_{t+1} = x_t + \eta \nabla \log p(x_t) + \sqrt{2\eta} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I), \quad (19)$$

for SGLD [1], and

$$v_{t+1} = \beta v_t + \eta \nabla \log p(x_t) + \sqrt{2\alpha\eta} \xi_t, \quad x_{t+1} = x_t + v_{t+1}, \quad (20)$$

for an SGHMC-style baseline [2]. Here η is the step size, α is a friction/noise parameter, and $\beta \in (0, 1)$ is momentum decay.

For particle methods, we use $N = 128$ particles; for single-chain methods, we compute metrics from a rolling window of recent iterates to ob-

tain comparable finite-sample summaries. Alongside wall-clock time, we report gradient- and kernel-evaluation counts as hardware-agnostic measures of computational cost.

All benchmarks use checkpointed evaluation at a benchmark-specific interval and a patience-based early-stopping rule triggered by persistent degradation of a designated monitor metric. Pure sampling benchmarks use KSD as the monitor (50D Gaussian, 2D synthetic targets, and Mixture2D), whereas predictive benchmarks use held-out NLL (UCI logistic regression, BNN, and HLR). For each run, we retain the best finite checkpoint under that monitor before stopping. Accuracy and ECE are reported for predictive tasks but are never used as stopping triggers. Benchmarks with pronounced numerical instability add a short non-finite fail-fast guard; we note those cases locally.

Across the full benchmark suite, the recurring pattern is that multirate SVGD variants improve robustness relative to vanilla SVGD, with the largest gains on anisotropic, hierarchical, and multimodal targets. Fixed multirate SVGD serves as a cheaper robustness baseline, while adaptive multirate SVGD usually delivers the strongest quality-cost tradeoff when local stiffness varies along the trajectory.

3.3. 50D Gaussian

This benchmark stresses stability in a high-dimensional setting with anisotropic covariance structure. We run 1,000 iterations with checkpointing every 50 steps and a 5-seed ablation. Reported metrics are moment errors ($\|\hat{\mu} - \mu\|_2$ and $\|\hat{\Sigma} - \Sigma\|_F$), ESS, KSD, and mean log probability.

Among the particle methods, Adapt-MR-SVGd is the only variant that keeps both moment errors and KSD under control in this anisotropic Gaus-

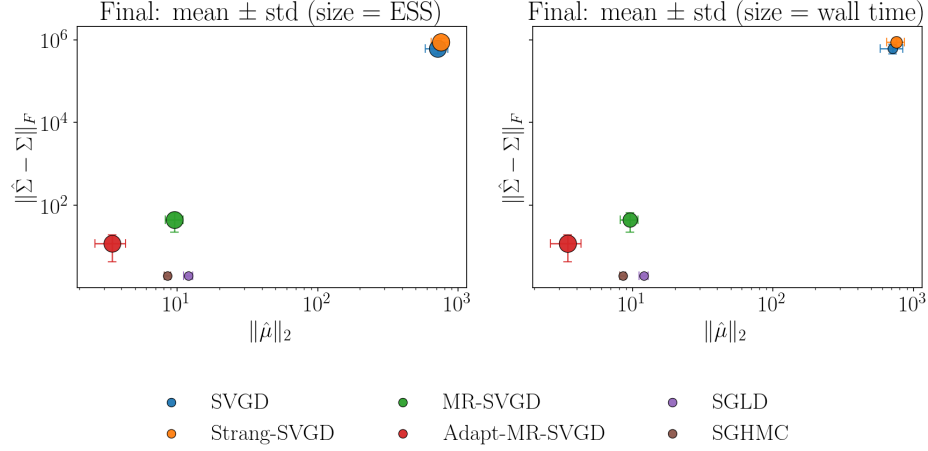


Figure 1: 50D Gaussian: final mean $\pm$ std Pareto plot in moment-error space with marker size encoding ESS (left) and wall time (right).

252 sian setting: it substantially improves on MR-SVGD and avoids the severe
 253 degradation exhibited by vanilla SVGD and Strang-SVGD. The chain base-
 254 lines remain competitive on some final-checkpoint fidelity metrics, but they
 255 do so with ESS near 4, far below the particle methods. The main lesson of
 256 this benchmark is therefore numerical rather than absolute: multirate treat-
 257 ment is critical for maintaining stable, informative particle behavior once
 258 drift and repulsion interact through strong anisotropy. Within the particle
 259 family, MR-SVGD offers a cheaper but less accurate fixed-schedule baseline,
 260 whereas Adapt-MR-SVGD achieves the strongest quality at a higher compu-
 261 tational cost.

262 3.4. 2D synthetic targets

263 We evaluate banana, ring, squiggly, two-moons, and funnel targets over
 264 five seeds, with 1,000 iterations and checkpointing every 20 steps. Here KSD
 265 serves both as the stop monitor and the primary quality metric, and we add

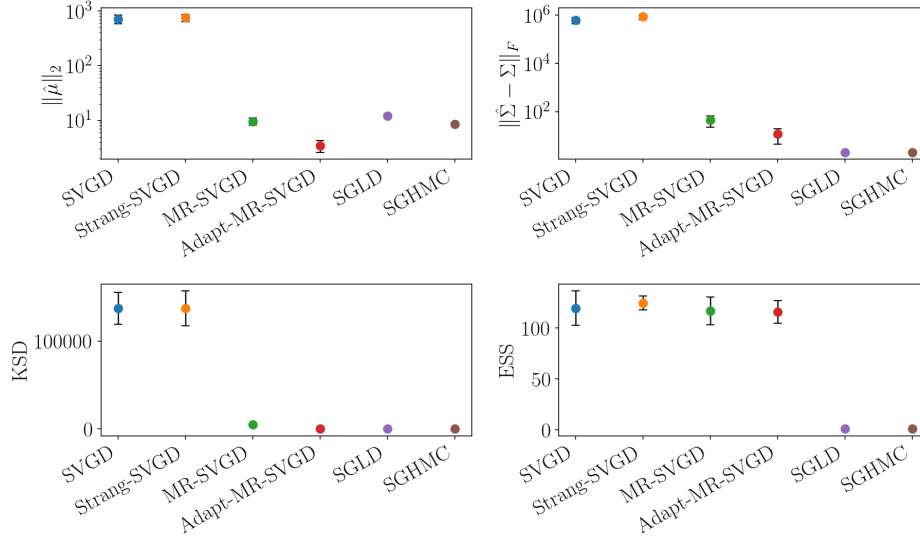
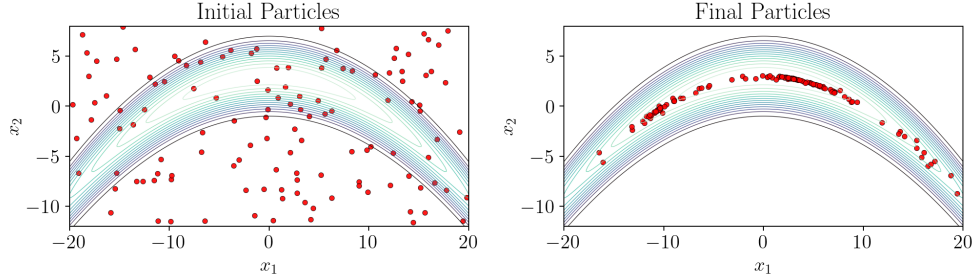


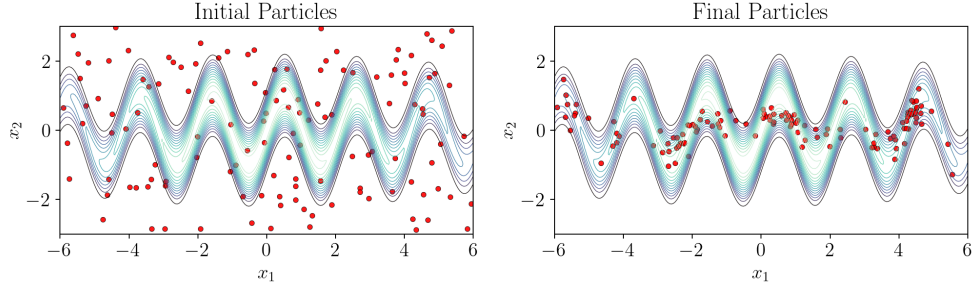
Figure 2: 50D Gaussian final-checkpoint summary across methods. The four panels report $\|\hat{\mu} - \mu\|_2$, $\|\hat{\Sigma} - \Sigma\|_F$, KSD, and ESS, respectively; markers show mean values across seeds and error bars indicate one standard deviation. Lower is better for the first three metrics, whereas higher is better for ESS. This figure highlights robustness differences under strong anisotropy: Adapt-MR-SVGD is the only particle variant that keeps both moment errors and KSD under control.

266 a short non-finite KSD fail-fast guard because several targets can destabi-
 267 lize weak particle updates. We report mean log probability as a secondary
 268 density-concentration score, gradient/kernel evaluations and wall time as cost
 269 metrics, and ESS as a supplemental mixing proxy. Moment errors and grid-
 270 based L_1 discrepancy are retained as diagnostics, but they are not used in
 271 the main 2D summary table. To contextualize the geometry of these targets,
 272 we also show short-run visualization panels for representative cases.

273 Across the full 2D suite, Adapt-MR-SVGD gives the strongest overall
 274 distributional fidelity: it is the only particle method with zero failures,



(a) Banana target: initial and short-run final particle states.

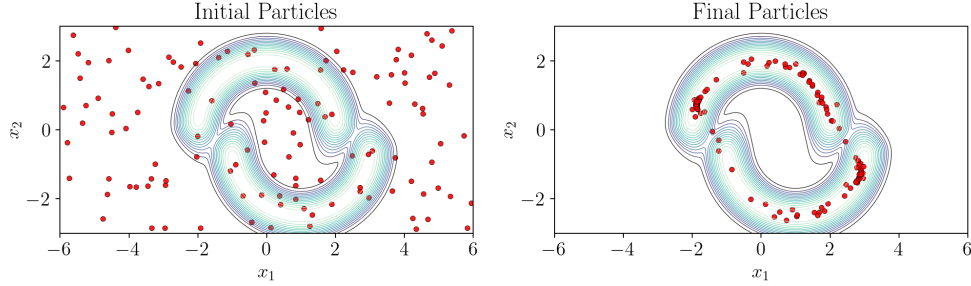


(b) Squiggly target: initial and short-run final particle states.

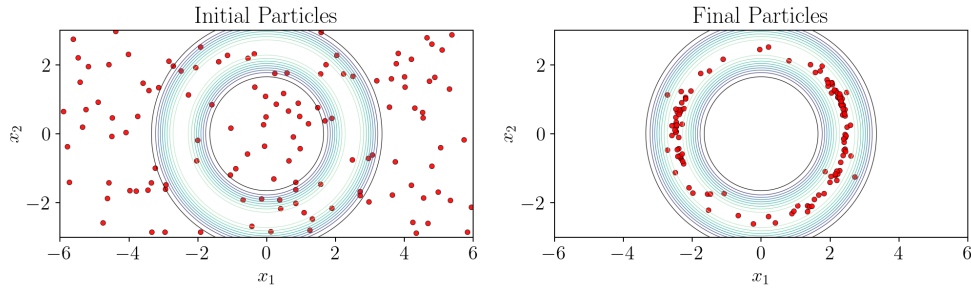
Figure 3: 2D target visualization panels produced with Adapt-MR-SVGD under visualization-only settings. Each subfigure shows target-density contours with initial particles (left) and short-run final particles (right).

275 attains the best overall median KSD, and also attains the best median
 276 mean log-density while retaining high ESS. The gain is concentrated on the
 277 geometry-driven ring, squiggly, and two-moons targets, where Adapt-MR-
 278 SVGD achieves the best mean KSD over seeds (approximately 0.231, 1.03,
 279 and 0.308, respectively) and the single-rate particle methods degrade sharply.
 280 On the easier banana target, by contrast, all particle methods attain nearly
 281 identical KSD values around 0.116, indicating that multirate treatment is
 282 not especially beneficial when the target geometry is comparatively mild.

283 The funnel target remains the main exception. Fixed MR-SVGD attains



(c) Two-moons target: initial and short-run final particle states.



(d) Ring target: initial and short-run final particle states.

Figure 3: 2D target visualization panels (continued).

284 the best mean KSD on funnel (0.261 versus 0.456 for Adapt-MR-SVGD),
 285 but Adapt-MR-SVGD still remains fully stable and yields substantially bet-
 286 ter mean log-density (-0.023 versus -2.89). Vanilla SVGD and Strang-
 287 SVGD each fail on four of the five funnel seeds, so finite-run medians alone
 288 would understate their instability; this is why table 1 reports failure counts
 289 alongside the quality and cost summaries. Overall, these 2D results sharpen
 290 the main qualitative claim: adaptive multirate integration is most valuable
 291 when the interaction between drift and repulsion becomes strongly geometry-
 292 dependent, rather than as a uniform improvement on every benign target. In
 293 this setting, ESS is much less discriminative than KSD and mean log-density,
 294 so we treat it as a supplemental mixing indicator rather than a primary qual-

Method	Fails↓	KSD↓	mean logp↑	ESS↑	grad evals↓	wall (s)↓
Adapt-MR-SVGD	0	0.308	-0.175	128.0	182	0.79
MR-SVGD	0	40.13	-278.0	121.8	80	1.13
SVGD	4	26.02	-1.93×10^4	128.0	80	0.32
Strang-SVGD	4	30.09	-1.20×10^5	121.7	121	1.55

Table 1: 2D targets quality-cost summary across particle methods. Entries are medians over the 25 target-seed runs, using the best checkpoint selected by the KSD monitor for each run. “Fails” counts runs whose KSD monitor never attained a finite value; methods are ordered first by failure count and then by median KSD. KSD is the primary distributional quality metric; mean log probability is reported as a secondary score, ESS as a supplemental mixing indicator, and gradient evaluations / wall time as cost metrics. Moment errors and grid L_1 remain diagnostic quantities and are omitted from this main summary. The main pattern is that Adapt-MR-SVGD is the only zero-failure particle method and gives the best overall KSD across the suite, while banana shows little separation and funnel remains a nuanced exception.

ity metric.

3.5. *Mixture2D (mix8)*

This benchmark emphasizes multimodal exploration and mass allocation across modes. All methods start from the same small Gaussian centered at the origin, away from the modal ring, and then run for a fixed budget of 1,000 iterations with checkpointing every 20 steps across 5 seeds. For this benchmark we use a shared multiscale RBF kernel with bandwidth factors $\{0.5, 1, 2\}$ for all particle methods so that local spacing and longer-range repulsion are both represented on the modal ring. We treat this as an experimental kernel-choice ablation rather than a change to the multirate update itself; prior SVGD work has also considered broader kernel families beyond

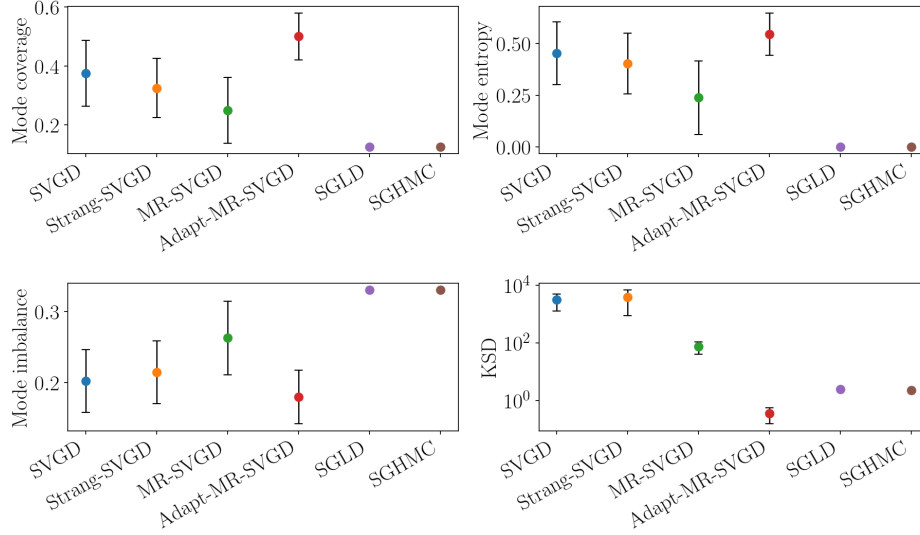


Figure 4: Mixture2D (mix8) fixed-budget final-checkpoint summary across methods. The four panels report mode coverage, mode entropy, mode imbalance, and KSD; markers show mean values across seeds and error bars indicate one standard deviation. Higher is better for coverage and entropy, while lower is better for mode imbalance and KSD.

306 a single scalar RBF [5, 21]. We define mode coverage by assigning each sam-
 307 ple to its nearest mixture center and counting a mode as covered once it
 308 carries at least 5% of the empirical mass. The primary metrics are mode
 309 coverage, normalized mode entropy, and mode imbalance; KSD and mean
 310 log probability are retained as secondary diagnostics, with ESS reported as
 311 a supplemental mixing proxy. Here imbalance is computed from the same
 312 nearest-center empirical mode masses and measures deviation from uniform
 313 mass allocation across the eight modes.

314 Under this harder off-mode initialization, adaptive MR-SVGD still gives
 315 the strongest overall multimodal exploration at the fixed budget, now reach-
 316 ing the highest mean final mode coverage and mode entropy across seeds

Table 2: Mixture2D (mix8) mean final metrics over five seeds at the fixed 1,000-iteration budget. Higher is better for coverage and entropy; lower is better for KSD. We omit minimum mass because it is zero for all methods at this budget. This table summarizes the fixed-budget multimodal result: Adapt-MR-SVGD gives the strongest overall exploration, while vanilla and Strang spread more than MR-SVGD but remain far less stable by KSD.

Method	Coverage $\uparrow$	Entropy $\uparrow$	KSD $\downarrow$
Adapt-MR-SVGD	0.500	0.544	0.360
SVGD	0.375	0.453	3162.9
Strang-SVGD	0.325	0.403	3883.0
MR-SVGD	0.250	0.238	74.3
SGHMC	0.125	0.000	2.29
SGLD	0.125	0.000	2.53

317 (0.500 and 0.544, respectively); table 2 provides the corresponding mean-
 318 over-seeds summary, while fig. 4 carries the cross-seed variability. The shared
 319 multiscale kernel also makes the single-rate particle baselines more compet-
 320 itive than under the plain single-RBF setting: vanilla SVGD and Strang-
 321 SVGD both cover more modes and attain higher mode entropy, but their
 322 KSD values remain far worse than the adaptive method, so the extra spread
 323 still comes with much weaker stability. Fixed MR-SVGD remains stable and
 324 slightly improves its KSD over the single-RBF version, but it does not match
 325 the adaptive variant on the main multimodal metrics. The imbalance panel
 326 in fig. 4 complements coverage and entropy by showing that the adaptive
 327 method also allocates mass more evenly across the discovered modes. SGLD
 328 and SGHMC remain trapped in a single mode, with final coverage fixed at

0.125 and zero mode entropy. The minimum mass statistic is still zero for all methods at this budget, so in this redesigned benchmark coverage, entropy, and imbalance are the informative multimodal diagnostics while KSD remains a secondary stability measure.

3.6. UCI logistic regression

We benchmark Bayesian logistic regression on `breast_cancer`, `ionosphere`, `spambase`, and `a5a`, using a Gaussian prior on weights (as implemented in the benchmark script) and 5 seeds per dataset. We report test accuracy, NLL, ECE, ESS, and mean log probability; KSD is optional and disabled by default in this benchmark configuration. Results are summarized with per-dataset panels in fig. 5 and a grouped best-NLL summary table (particle methods only) in table 3; both use the best finite-NLL checkpoint selected under the global predictive stopping policy [22].

UCI logistic regression serves mainly as a predictive sanity check rather than as the flagship evidence for multirate gains. The rankings are mixed. Adapt-MR-SVGD gives the best overall NLL on `breast_cancer`, while on `ionosphere`, `spambase`, and `a5a` either SGHMC or a single-rate particle method is competitive or superior. This weaker separation is consistent with the main claim of the paper: multirate benefits are most pronounced on anisotropic, multimodal, or hierarchical posteriors, whereas lower-dimensional logistic regression tasks show smaller and less uniform gains. On the smaller datasets, accuracy is also tightly quantized, so NLL is the more discriminative summary.

352 3.7. Bayesian neural network

353 We evaluate a one-hidden-layer BNN (hidden width 32) on spambase
354 and a5a, using a Gaussian prior, a fixed split seed, and 5 sampler seeds per
355 dataset. For each dataset, we carve a validation split from the training data,
356 select checkpoints by validation NLL under the global predictive stopping
357 policy, and report test accuracy, test NLL, test ECE, ESS, and mean log
358 probability at the selected checkpoint. KSD is optional and disabled by
359 default in this configuration.

360 The two BNN datasets give a more informative but less uniform predictive
361 story than the earlier small-data configuration. On spambase, Adapt-MR-
362 SVGD attains the best overall test NLL (0.174) and the highest mean test
363 accuracy (0.940), with the remaining particle methods clustered together
364 around 0.187. On a5a, the ranking shifts: MR-SVGD attains the best overall
365 test NLL (0.334) and the highest mean test accuracy (0.844), Adapt-MR-
366 SVGD remains the second-best particle method (0.347), and the single-rate
367 particle variants are again worse. The chain baselines are competitive in
368 isolated cases—SGHMC on spambase and SGLD on a5a—but the multirate
369 particle methods form the strongest pair overall across the two datasets.
370 This makes the BNN benchmark corroborative rather than primary evidence:
371 multirate structure improves predictive quality over the single-rate particle
372 baselines, but the preferred multirate variant depends on the dataset.

373 3.8. Large-scale hierarchical logistic regression

374 To stress-test multirate behavior in a high-dimensional anisotropic pos-
375 terior, we use a large-scale hierarchical logistic regression benchmark with
376 grouped random effects. The model combines high parameter dimension

377 with funnel-like geometry induced by a global scale parameter, making it a
 378 natural benchmark for joint quality-cost assessment of SVGD variants.

379 *Problem definition.* Given binary outcomes $y_i \in \{0, 1\}$, sparse features $x_i \in$
 380 $\mathbb{R}^p$, and group indices $g_i \in \{1, \dots, G\}$, we define

$$\begin{aligned} y_i \mid x_i, g_i, \beta, \alpha, u &\sim \text{Bernoulli}(\sigma(\alpha + x_i^\top \beta + u_{g_i})), \\ i &= 1, \dots, n, \end{aligned} \quad (21)$$

381 where $\sigma(\cdot)$ is the logistic link. We use a non-centered parameterization for
 382 group effects,

$$\begin{aligned} \beta &\sim \mathcal{N}(0, \sigma_\beta^2 I_p), \quad \alpha \sim \mathcal{N}(0, \sigma_\alpha^2), \\ u_j &= \tau z_j, \quad z_j \sim \mathcal{N}(0, 1), \\ \log \tau &\sim \mathcal{N}(\mu_\tau, \sigma_\tau^2), \end{aligned} \quad (22)$$

383 for $j = 1, \dots, G$. The posterior (up to normalization) is

$$\begin{aligned} \pi(\theta) &\propto \left[\prod_{i=1}^n \text{Bernoulli}(y_i; \sigma(\alpha + x_i^\top \beta + u_{g_i})) \right] \\ &\quad \times p(\beta) p(\alpha) p(z) p(\log \tau), \\ \theta &= (\beta, \alpha, z, \log \tau). \end{aligned} \quad (23)$$

384 *Protocol.* For the long-tail grouped setting, we run $n = 10^6$ observations
 385 with $p = 300$ features and $G = 50,000$ groups. Each method runs up to
 386 1,000 iterations with checkpointing every 20 iterations and 5 random seeds;
 387 particle methods use 32 particles, and single-chain baselines use a rolling-
 388 window evaluation set. This benchmark follows the global predictive NLL-
 389 based stopping policy and adds a two-checkpoint non-finite fail-fast guard
 390 because unstable methods can quickly leave the finite-NLL regime. ECE is
 391 reported as a metric but is not used as a stopping trigger.

392 *Result summary.* table 5 shows that Adapt-MR-SVGD achieves the best
 393 mean best-NLL while remaining finite across all seeds. MR-SVGD is a close
 394 second in NLL and reaches its best checkpoint faster on average (92.5 s ver-
 395 sus 123.3 s). SVGD/Strang-SVGD and SGHMC show frequent non-finite
 396 behavior in this regime, while SGLD remains finite more often but with sub-
 397 stantially worse NLL.

398 *Uniform-group sensitivity check.* table 6 provides a compact robustness check
 399 under uniform group assignments. Adapt-MR-SVGD remains best in mean
 400 best-NLL with full finite-seed coverage (5/5), and MR-SVGD remains close
 401 while being fastest among methods with full finite coverage. This supports
 402 the same ordering seen in the long-tail regime while keeping the long-tail
 403 setting as the primary stress test for our main claim.

404 4. Conclusions

405 This work studies a central numerical weakness of SVGD: a single global
 406 step size must simultaneously resolve attractive drift and diversity-preserving
 407 repulsion even when these components evolve on different effective time
 408 scales. We address this mismatch by recasting SVGD as a split interacting-
 409 particle flow and integrating the drift and repulsion terms with multirate
 410 updates.

411 Across a broad benchmark suite, the main empirical pattern is that multi-
 412 rate SVGD improves robustness and quality-cost tradeoffs relative to vanilla
 413 SVGD, with the strongest gains on anisotropic, multimodal, and hierarchi-
 414 cal targets. Fixed multirate schedules provide a simpler robustness baseline,
 415 while adaptive drift resolution offers the strongest performance when local

416 stiffness changes along the trajectory. At the same time, no single vari-
417 ant dominates every metric and every regime, reinforcing the importance of
418 evaluating quality and cost jointly rather than through a single summary
419 statistic.

420 Several limitations remain. Current adaptivity is centered on drift reso-
421 lution, while repulsion frequency is still controlled by simple schedules; ESS
422 remains a univariate proxy; and dense kernel interactions continue to con-
423 strain scaling at large particle counts. These observations suggest clear
424 next steps: tighter adaptive controllers that coordinate drift and repul-
425 sion updates, approximate or structured kernel computations for large- N
426 regimes, and stronger theoretical links between multirate stability analysis
427 and particle-based Bayesian inference guarantees. Together, these directions
428 can move multirate SVGD from a robust empirical strategy toward a more
429 principled and scalable foundation for modern Bayesian computation.

430 Data availability

431 Code, experiment scripts, and manuscript sources for this study are main-
432 tained in a public GitHub repository.¹ Public benchmark inputs used in the
433 predictive experiments are derived from standard UCI datasets; all other
434 benchmark targets are synthetic and are generated directly by the repository
435 code. Generated figures, metrics, and supplementary animation artifacts are
436 available in the project repository for reference and reproduction.

¹<https://github.com/csml-beach/multirate-sampling>

437 Declaration of generative AI and AI-assisted technologies

438 The jax codebase for this project was developed in part with the assistance
439 of OpenAI Codex. After using this tool/service, the author reviewed and
440 edited the content as needed and takes full responsibility for the content of
441 the published article.

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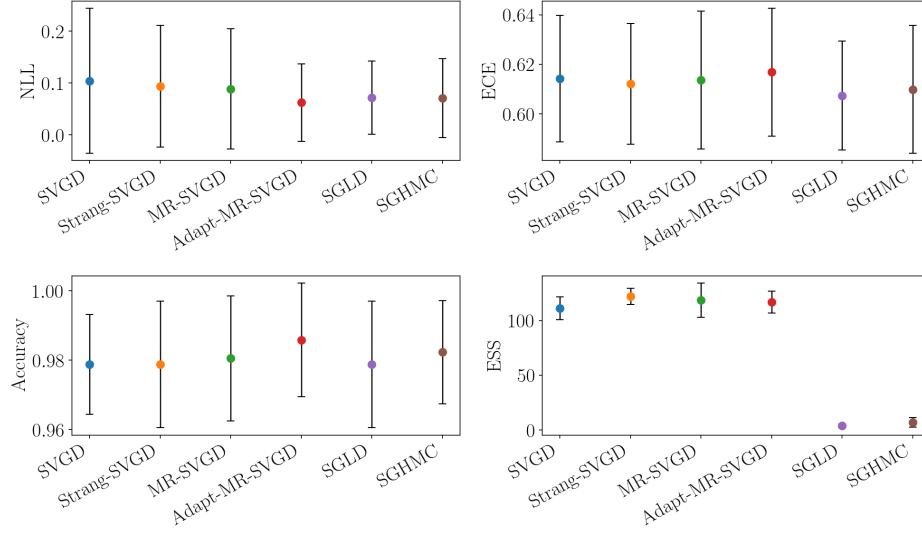
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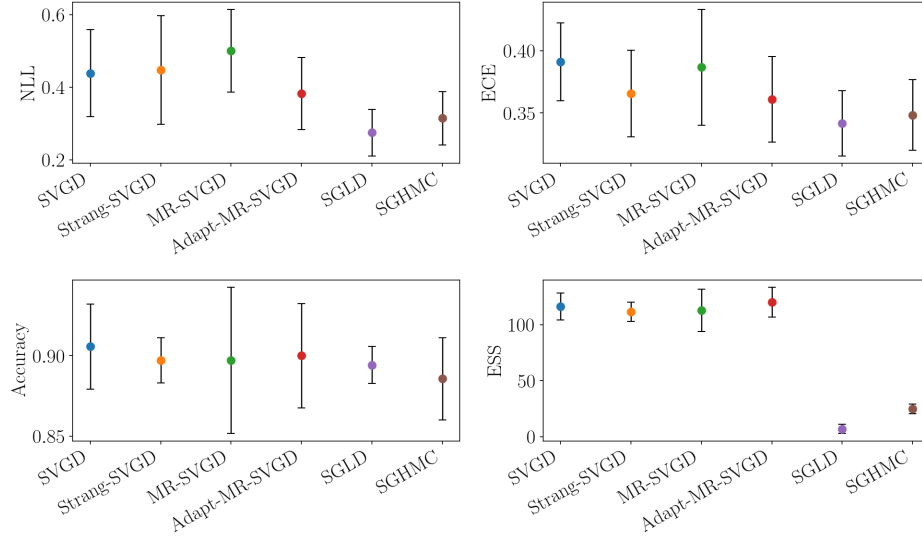
Algorithm 2 Adaptive multirate SVGD (Adapt-MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^N$, step size h , bounds $m_{\min}, m_{\max}$, tolerance Tol

```
1:  $x \leftarrow \{x_i\}$ 
2: compute repulsion  $f_{\text{rep}}(x)$  using eq. (5)
3:  $x \leftarrow x + h f_{\text{rep}}(x)$ 
4: compute drift  $f_{\text{drift}}(x)$  using eq. (4)
5:  $x_{\text{full}} \leftarrow x + h f_{\text{drift}}(x)$ 
6:  $x_{\text{half}} \leftarrow x + (h/2) f_{\text{drift}}(x)$ 
7: compute drift  $f_{\text{drift}}(x_{\text{half}})$ 
8:  $\tilde{x}_{\text{full}} \leftarrow x_{\text{half}} + (h/2) f_{\text{drift}}(x_{\text{half}})$ 
9: estimate relative error  $\rho = \|\tilde{x}_{\text{full}} - x_{\text{full}}\|/(\|x\| + \epsilon)$ 
10:  $m \leftarrow \text{clip}(\lceil \sqrt{\rho/\text{Tol}} \rceil, m_{\min}, m_{\max})$ 
11: if  $m \leq 2$  then
12:    $x \leftarrow \tilde{x}_{\text{full}}$   $\triangleright$  reuse current predictor if accurate
13: else
14:   for  $s = 1$  to  $m$  do
15:     compute drift  $f_{\text{drift}}(x)$ 
16:      $x \leftarrow x + (h/m) f_{\text{drift}}(x)$ 
17:   end for
18: end if
19: return  $x$ 
```

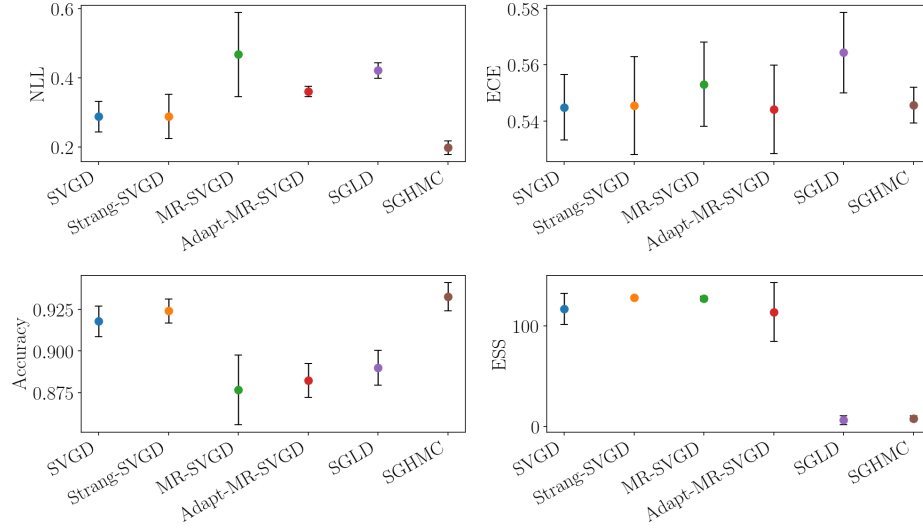


(a) breast_cancer

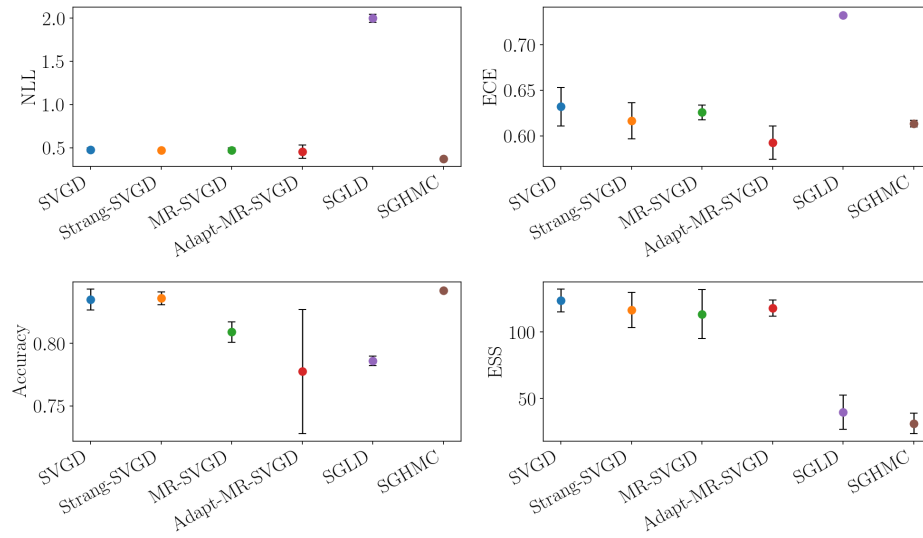


(b) ionosphere

Figure 5: UCI logistic regression summary across datasets. Each subpanel reports the best-checkpoint mean across seeds with one standard deviation for test accuracy, NLL, ECE, and ESS, where the checkpoint is selected by the NLL-based early-stop monitor. Higher is better for accuracy and ESS; lower is better for NLL and ECE. Remaining datasets are shown in the continued panel.



(c) spambase

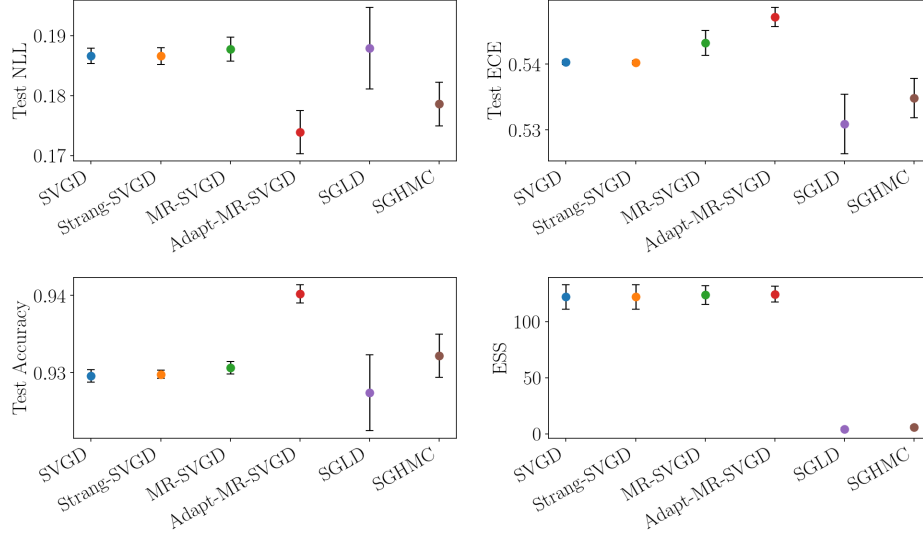


(d) a5a

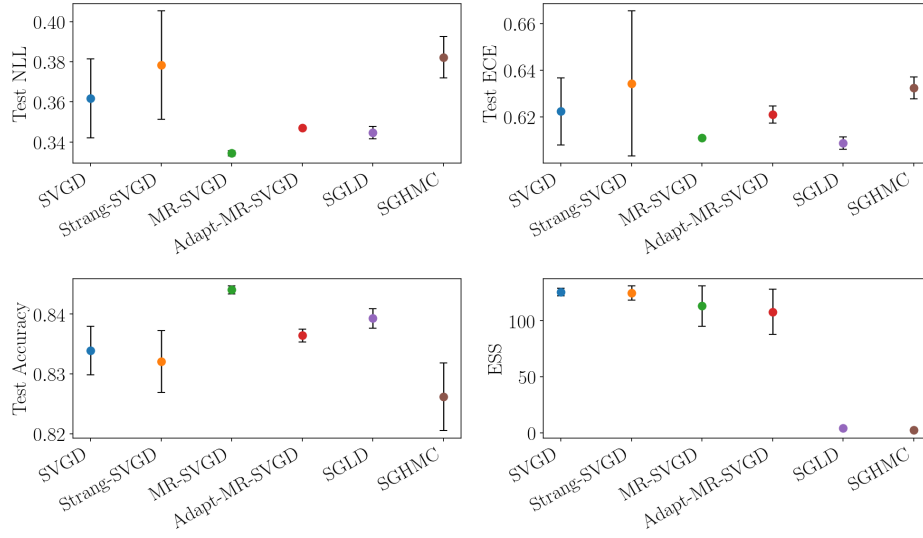
Figure 5: UCI logistic regression summary across datasets (continued).

Method	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
<i>breast_cancer</i>			
Adapt-MR-SVGD	0.062	0.986	8.9
MR-SVGD	0.089	0.981	6.2
Strang-SVGD	0.094	0.979	6.5
SVGD	0.104	0.979	2.0
<i>ionosphere</i>			
Adapt-MR-SVGD	0.383	0.900	3.4
SVGD	0.438	0.906	0.9
Strang-SVGD	0.447	0.897	5.2
MR-SVGD	0.500	0.897	6.1
<i>spambase</i>			
SVGD	0.288	0.918	1.3
Strang-SVGD	0.288	0.924	8.8
Adapt-MR-SVGD	0.361	0.882	8.1
MR-SVGD	0.467	0.877	4.3
<i>a5a</i>			
Adapt-MR-SVGD	0.453	0.778	36.5
Strang-SVGD	0.471	0.836	3.0
MR-SVGD	0.474	0.809	14.6
SVGD	0.478	0.835	2.6

Table 3: UCI logistic-regression particle-method summary over five seeds. For each dataset and method, we select the best finite-NLL checkpoint per seed and report the method-wise mean of that best NLL, the accuracy at that checkpoint, and the corresponding wall time. ECE is summarized in the per-dataset panels of fig. 5.



(a) spambase



(b) a5a

Figure 6: BNN summary across datasets. Each subpanel reports the best-checkpoint mean across seeds with one standard deviation for test accuracy, NLL, ECE, and ESS, where the checkpoint is selected by the validation-NLL early-stop monitor. Higher is better for accuracy and ESS; lower is better for NLL and ECE.

Method	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
<i>spambase</i>			
Adapt-MR-SVGD	0.174	0.940	35.7
Strang-SVGD	0.187	0.930	51.2
SVGD	0.187	0.930	21.1
MR-SVGD	0.188	0.931	51.5
<i>a5a</i>			
MR-SVGD	0.334	0.844	5.3
Adapt-MR-SVGD	0.347	0.836	8.1
SVGD	0.362	0.834	17.5
Strang-SVGD	0.378	0.832	49.6

Table 4: BNN particle-method summary over five seeds. For each dataset and method, we select the best finite-validation-NLL checkpoint per seed and report the method-wise mean of the corresponding test NLL, test accuracy, and wall time. ECE is summarized in the per-dataset panels of fig. 6.

Method	Finite-best↑	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
Adapt-MR-SVGD	5/5	0.613	0.658	123.3
MR-SVGD	5/5	0.623	0.656	92.5
SGHMC	2/5	0.860	0.560	77.9
SVGD	3/5	0.885	0.577	154.2
Strang-SVGD	3/5	1.263	0.579	163.6
SGLD	4/5	1.823	0.576	101.6

Table 5: HLR long-tail summary over five seeds. For each seed and method, we select the best finite-NLL checkpoint; the table reports method-wise means of that best NLL, the accuracy at that checkpoint, and the corresponding wall time. “Seeds with finite best” counts runs that reached at least one finite-NLL checkpoint before stopping. This table is the main hierarchical stress test: Adapt-MR-SVGD achieves the best best-NLL while remaining finite on all seeds, and MR-SVGD is the closest lower-cost baseline.

Method	Finite-best $\uparrow$	Best NLL $\downarrow$	Acc@best NLL $\uparrow$	Wall@best (s) $\downarrow$
Adapt-MR-SVGD	5/5	0.681	0.601	94.6
SVGD	1/5	0.700	0.548	54.9
MR-SVGD	5/5	0.706	0.597	44.2
Strang-SVGD	1/5	0.708	0.584	241.7
SGLD	5/5	0.743	0.528	66.0
SGHMC	5/5	0.791	0.554	79.7

Table 6: HLR uniform-group summary over five seeds. As in table 5, each entry uses the best finite-NLL checkpoint per seed and reports method-wise means for NLL, accuracy at that checkpoint, and wall time.